Gerrymandering Metrics

2024 Election Results and the 2024 District Map

Part 4

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0
──
✔ dplyr     1.1.4     ✔ readr     2.1.5
✔ forcats   1.0.1     ✔ stringr   1.5.2
✔ ggplot2   4.0.0     ✔ tibble    3.3.0
✔ lubridate 1.9.4     ✔ tidyr     1.3.1
✔ purrr     1.1.0
── Conflicts ────────────────────────────────────────── tidyverse_conflicts()
──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
sv <- read_csv("data/clean/g24_svprec_clean.csv")
```

```
Rows: 51123 Columns: 76
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
chr (29): FIPS, SVPREC, SVPREC_KEY, ELECTION, GEO_TYPE, ASSAIP01,
ASSDEM01, ...
dbl (47): COUNTY, ADDIST, CDDIST, SDDIST, BEDIST, TOTREG, DEMREG, REPREG,
AI...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

```
compute_metrics <- function(df, votesA = "DEMVOTE", votesB = "REPVOTE") {
df2 <- df %>%
mutate(votes_A = .data[[votesA]],
votes_B = .data[[votesB]],
```

```r
total = votes_A + votes_B,
share_A = votes_A / total,
share_B = votes_B / total,
winner = if_else(votes_A > votes_B, "A", "B"),
wasted_A = if_else(winner == "A", votes_A - (total/2 + 0.5), votes_A),
wasted_B = if_else(winner == "B", votes_B - (total/2 + 0.5), votes_B))
mean_share <- mean(df2$share_A, na.rm = TRUE)
median_share <- median(df2$share_A, na.rm = TRUE)
mean_median <- mean_share - median_share
total_wasted_A <- sum(df2$wasted_A, na.rm = TRUE)
total_wasted_B <- sum(df2$wasted_B, na.rm = TRUE)
efficiency_gap <- (total_wasted_A - total_wasted_B) / sum(df2$total, na.rm =
TRUE)
list(mean_median = mean_median, efficiency_gap = efficiency_gap, seats_A =
sum(df2$winner == "A", na.rm = TRUE), seats_B = sum(df2$winner == "B", na.rm =
TRUE))
}
sv <- read_csv("data/clean/g24_svprec_clean.csv")
```

```
Rows: 51123 Columns: 76
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
chr (29): FIPS, SVPREC, SVPREC_KEY, ELECTION, GEO_TYPE, ASSAIP01,
ASSDEM01, ...
dbl (47): COUNTY, ADDIST, CDDIST, SDDIST, BEDIST, TOTREG, DEMREG, REPREG,
AI...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

```r
district_votes <- sv %>%
group_by(CDDIST) %>%
summarise(DEMVOTE = sum(DEMVOTE, na.rm = TRUE),
REPVOTE = sum(REPVOTE, na.rm = TRUE))
metrics_2024 <- compute_metrics(district_votes)
metrics_2024
```

```
$mean_median
[1] NA

$efficiency_gap
[1] Inf
```

```
$seats_A
[1] 0

$seats_B
[1] 53
```

```r
district_votes <- sv %>%
  group_by(CDDIST) %>%
  summarise(
    DEMVOTE = sum(as.numeric(CNGDEM01), na.rm = TRUE) +
              sum(as.numeric(CNGDEM02), na.rm = TRUE),

    REPVOTE = sum(as.numeric(CNGREP01), na.rm = TRUE) +
              sum(as.numeric(CNGREP02), na.rm = TRUE)
  )
```

```
Warning: There were 208 warnings in `summarise()`.
The first warning was:
ℹ In argument: `DEMVOTE = +...`.
ℹ In group 2: `CDDIST = 1`.
Caused by warning:
! NAs introduced by coercion
ℹ Run `dplyr::last_dplyr_warnings()` to see the 207 remaining warnings.
```

```r
print(district_votes)
```

```
# A tibble: 53 × 3
   CDDIST  DEMVOTE  REPVOTE
    <dbl>    <dbl>    <dbl>
 1      0 54707782 35374877
 2      1   110472   208150
 3      2   272384   106407
 4      3   187960   233895
 5      4   227321   114644
 6      5   134467   214223
 7      6   165386   121625
 8      7   197361    98273
 9      8   201756    70932
10      9   130093   121006
# ℹ 43 more rows
```

```r
district_votes_contested <- district_votes %>%
  filter((DEMVOTE + REPVOTE) > 0)
```

```r
metrics_2024 <- compute_metrics(district_votes_contested)
print(metrics_2024)
```

```
$mean_median
[1] -0.01162013

$efficiency_gap
[1] -0.2576804

$seats_A
[1] 44

$seats_B
[1] 9
```

Part 6: Calculating Gerrymandering Again for ab604_est

```r
library(tidyverse)
ab604 <- read_csv("data/derived/ab604_estimated_votes_area_weighted.csv")
```

```
Rows: 52 Columns: 3
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
chr (1): ab604_district
dbl (2): DEMVOTE, REPVOTE

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

```r
compute_newmetrics <- function(df, votesA = "DEMVOTE", votesB = "REPVOTE") {
  district_stats <- df %>%
    mutate( votes_A = .data[[votesA]], votes_B = .data[[votesB]], total_votes
= votes_A + votes_B) %>%
    filter(total_votes > 0) %>%
    mutate(share_A = votes_A / total_votes, share_B = votes_B / total_votes,
winner = if_else(votes_A > votes_B, "A", "B"), votes_needed_to_win =
floor(total_votes / 2) + 1,
      wasted_A = if_else(
        winner == "A", votes_A - votes_needed_to_win,
        votes_A), wasted_B = if_else(
        winner == "B", votes_B - votes_needed_to_win, votes_B))
  mean_share  <- mean(district_stats$share_A, na.rm = TRUE)
  median_share <- median(district_stats$share_A, na.rm = TRUE)
```

```r
  mean_median_gap <- mean_share - median_share
  total_wasted_A <- sum(district_stats$wasted_A, na.rm = TRUE)
  total_wasted_B <- sum(district_stats$wasted_B, na.rm = TRUE)
  total_votes_map <- sum(district_stats$total_votes, na.rm = TRUE)
  efficiency_gap <- (total_wasted_A - total_wasted_B) / total_votes_map
  seats_A <- sum(district_stats$winner == "A", na.rm = TRUE)
  seats_B <- sum(district_stats$winner == "B", na.rm = TRUE)
  list(
    mean_median = mean_median_gap,
    efficiency_gap = efficiency_gap,
    seats_A = seats_A,
    seats_B = seats_B
  )
}
metrics_ab604 <- compute_newmetrics(ab604)
print(metrics_ab604)
```

```
$mean_median
[1] 0.02760472

$efficiency_gap
[1] -0.1783588

$seats_A
[1] 46

$seats_B
[1] 6
```